

Dwiti Thaker

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Summary

Backend and AI Engineer with hands-on experience building scalable, distributed REST APIs and LLM-powered systems using Python. Specialise in RAG pipelines, agentic workflows, and high-performance backend services — with a track record of measurable improvements in production environments. Passionate about developer tooling and automation infrastructure.

Experience

Third Rock Techno | Python Backend & AI Intern

Aug 2025 – present

- Engineered and maintained scalable backend services using Python, building REST APIs and integrating AI frameworks to support internal product features and data workflows.
- Implemented validation, logging, and monitoring middleware for backend services, improving API reliability by 20% and reducing email verification failures by 40%.

Education and Certification

Indus University - B.Tech, Information Technology (Final Year) **CGPA: 9.34**

Technical Skills

- Languages:** Python, C, C++, SQL
- Backend:** FastAPI, REST APIs, JWT Authentication, OAuth2
- AI / LLM:** Retrieval-Augmented Generation (RAG), LangChain, LangGraph, LangSmith, LLM Agents, NLP, Embedding Models, Scikit-learn, Vector Databases
- Automation & Workflows:** n8n, AI Workflow Automation, API Integrations
- Data:** Pandas, NumPy, Matplotlib, Exploratory Data Analysis (EDA)
- Frontend:** React, Streamlit
- Tools:** Git, GitHub, VS Code, Jupyter Notebook, OpenAI API, Vercel, Docker
- Databases:** PostgreSQL, MongoDB, Supabase, SQL

Projects

🔗 Hybrid Retrieval-Augmented Generation (RAG) System | Python, LangChain, Qdrant, MongoDB

- Built a hybrid Retrieval-Augmented Generation (RAG) pipeline using BM25 and dense embeddings, improving retrieval precision by 32% across 24k+ documents.
- Reduced end-to-end query latency by ~30–40% and improved context relevance by ~20–25% using chunk-based indexing, score normalization, and hybrid rank fusion within a modular ML architecture.

🔗 Google Calendar Scheduling System | FastAPI, Supabase, Google Calendar API, FastMCP

- Engineered an async MCP-based scheduling backend integrating Google Calendar and Supabase, reducing average booking time by 10 minutes per session.
- Built fault-tolerant conflict resolution logic using FastMCP, automatically handling 7 weekly scheduling conflicts that previously required manual intervention raising booking success rate by 8%.

🔗 Reflexion Self-Improving AI Agent | Python, LangChain, LangGraph

- Built a self-improving AI agent using the Reflexion architecture (LangChain + LangGraph) with iterative self-evaluation loops, reducing solution time by 7 seconds vs a baseline single-pass agent.
- Implemented feedback-driven reasoning loops suitable for Copilot-style autonomous agents.

🔗 Hospital Management System | FastAPI, MongoDB

- Architected backend APIs with role-based access control (RBAC), validation workflows, and secure CRUD operations for patient and appointment management.

🔗 Anonymous Form Service | FastAPI, MongoDB

- Designed and deployed a secure anonymous feedback service using FastAPI and MongoDB, improving form submission success rate by 25% through strict schema validation and idempotent request handling. Deployed to Vercel via GitHub Actions.

Certification

- AI Programming with Python: Udacity